

KNN

Classification Algorithms

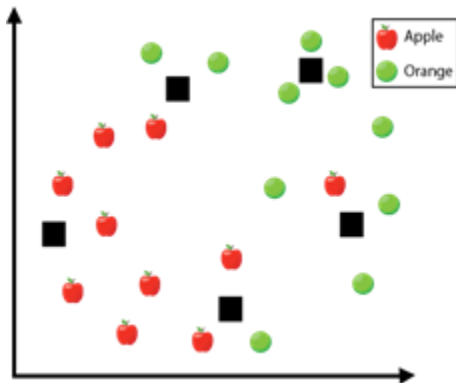
Formalized Folk Wisdom $\Rightarrow$ kNN

- ▶ “Ornithological species of identical plumage congregate concurrently”
- ▶ Argument by analogy
- ▶ Classify an unknown sample by looking at its “neighbors”

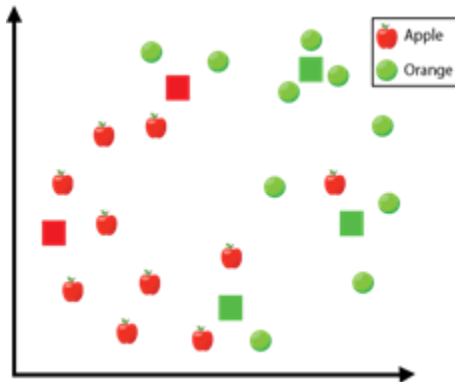
Goal: Classify Unknown Samples

- ▶ Two features
 - Only for plotting convenience
- ▶ Training: 10 RED + 10 GREEN
- ▶ Testing: 5 BLACK

Graphical View



But We Know!



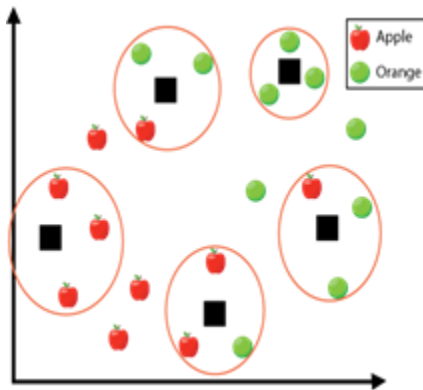
First Look at kNN

Recall the idea:

- ▶ Group by “nearness”

Method:

- ▶ Determine 3 nearest neighbors
- ▶ Assign majority label



What is the Accuracy?

- ▶ 3 out of 5 got correct:
 - Accuracy = 60%
 - Error = 40%
- ▶ Is it good or bad or ugly?

What is the Accuracy?

- ▶ 3 out of 5 got correct:
 - Accuracy = 60%
 - Error = 40%
- ▶ Is it good or bad or ugly?
- ▶ A random guess could have given 50%!!

- ▶ We “assumed” $k = 3$. It can also be 5, 7 . . . 10
 - Often odd. Why?
- ▶ The data can have many more classes
 - Apples, Oranges and Mangoes
- ▶ Distances need not be Euclidean.
 - Other distance functions exist

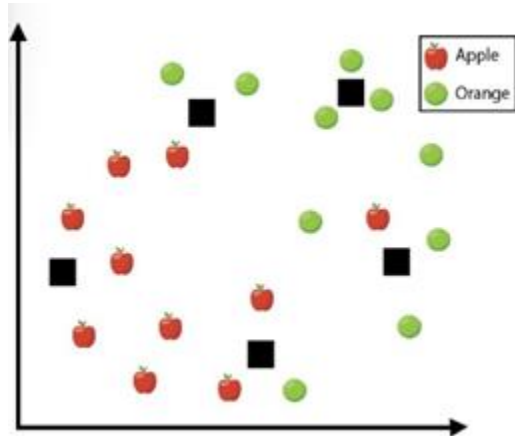
Will Another k Help? Some Times.

Idea:

- ▶ Try and pick the best

Method:

- ▶ Try $k = 5$
- ▶ What is the Accuracy?



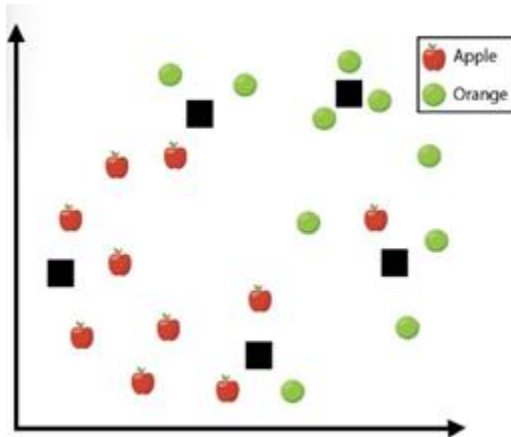
Will Another k Help? Some Times.

Idea:

- ▶ Try and pick the best

Method:

- ▶ Try $k = 5$
- ▶ What is the Accuracy?
 - 80%



k Nearest Neighbours (kNN)

Given k , Data and Distance Function

- ▶ Find the distance from z to all the samples in X
- ▶ Identify the k -Nearest Neighbours (smallest distances) and their class labels
- ▶ Classify z as the majority label from the k -Nearest Neighbours

Characteristics

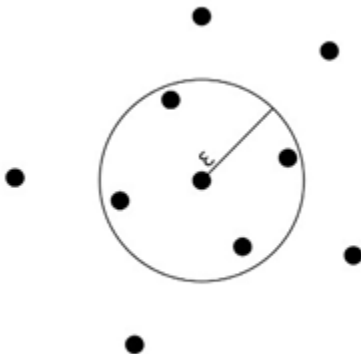
- ▶ KNN: n distance computations each of $O(d)$
 - number of samples = n
 - number of features or dimensionality = d
- ▶ Too many operations
- ▶ Needs all training samples at test
- ▶ time Storage intensive
- ▶ Computation intensive

- ▶ If a sample is not in the "K Neighbours" of at least one sample, it is an outlier



Epsilon Neighbourhood

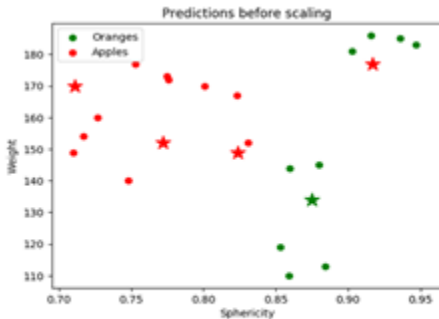
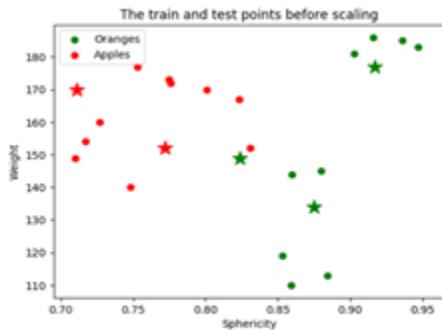
- Samples within a radius of ϵ from an sample



- ▶ Demo_KNN_Scaling
 - Dataset : Fruits
 - Objective : How characteristics of the data affect KNN effectiveness

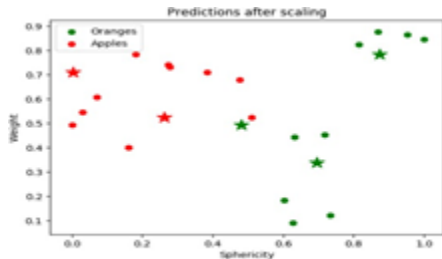
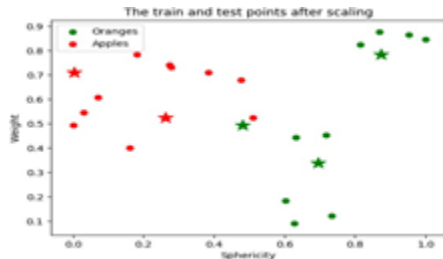
Scaling

How Scaling Affects the kNN?



Accuracy = 60%

How Scaling Affects the kNN?



Accuracy = 100%

There is a change in the accuracy before and after scaling the data.

Distance Functions

- ▶ Used to measure similarity between samples Different
- ▶ types in different contexts
- ▶ Examples:
 - Euclidean
 - Manhattan
 - Minkowski
 - Hamming

Distance Functions

- ▶ Euclidean:

$$\sqrt{\sum_{i=1}^k (x_i - x'_i)^2}$$

- ▶ Manhattan:

$$\sum_{i=1}^k |x_i - x'_i|$$

Distance Functions

- Minkowski:

$$\left(\sum_{i=1}^k (|x_i - x'_i|)^q \right)^{1/q}$$

- Hamming:

$$D_H = \sum_{i=1}^k |x_i - x'_i|$$

$$x = x' \Rightarrow D = 0$$

$$x \neq x' \Rightarrow D = 1$$

Experiments :

- ▶ Demo_KNN_Advertising_data
 - Dataset : Advertising
 - Objective : To classify the data using KNN classifier

Thanks!!
Questions?